

MODUL 12

Nama :Ewaldo Ardiansyah Widyadhana

Nim :109082500008

Kelas :IF-13-04

Soal1

```
package main

import "fmt"

func main() {
    var suara int
    var totalMasuk, totalSah int
    var hitungSuara [21]int

    for {
        fmt.Scan(&suara)
        if suara == 0 {
            break
        }
        totalMasuk++

        if suara >= 1 && suara <= 20 {
            totalSah++
            hitungSuara[suara]++
        }
    }
}
```

```

}

fmt.Printf("Suara masuk: %d\n", totalMasuk)

fmt.Printf("Suara sah: %d\n", totalSah)


for i := 1; i <= 20; i++ {

    if hitungSuara[i] > 0 {

        fmt.Printf("%d: %d\n", i, hitungSuara[i])

    }

}

}

```

Screenshot

The screenshot shows the Visual Studio Code interface with a Go project named 'laprakalpro2'. The Explorer panel on the left shows the project structure, including files like 'laprakalpro2', 'Modul10.go', 'Modul12soal1.go', 'package main.go', 'tempCodeRunnerFile.go', and 'TUGAS LAPRAK MODUL 10.pdf'. The main editor displays the code for 'Modul12soal1.go', which is a Go program that reads 20 integers from the user, counts the number of valid votes (between 1 and 20), and prints the total number of valid votes and the total number of votes entered.

```

1 package main
2
3 import "fmt"
4
5 func main() {
6     var suara int
7     var totalMasuk, totalSah int
8     var hitungSuara [21]int
9
10    for {
11        fmt.Scan(&suara)
12        if suara == 0 {
13            break
14        }
15        totalMasuk++
16
17        if suara >= 1 && suara <= 20 {
18            totalSah++
19            hitungSuara[suara]++
20        }
21    }

```

The terminal output shows the execution of the program, displaying the input votes and the final counts:

```

PS C:\laprakalpro2> go run "c:\laprakalpro2\modul12soal1.go"
7 19 3 2 72 1 -3 18 19 0
Suara masuk: 9
Suara sah: 7
1: 1
2: 1
3: 1
7: 1
18: 1
19: 2
PS C:\laprakalpro2>

```

Soal2

```
package main

import "fmt"

func main() {
    var suara int
    var totalMasuk, totalSah int
    var hitungSuara [21]int

    for {
        fmt.Scan(&suara)
        if suara == 0 {
            break
        }
        totalMasuk++

        if suara >= 1 && suara <= 20 {
            totalSah++
            hitungSuara[suara]++
        }
    }

    fmt.Printf("Suara masuk: %d\n", totalMasuk)
    fmt.Printf("Suara sah: %d\n", totalSah)
```

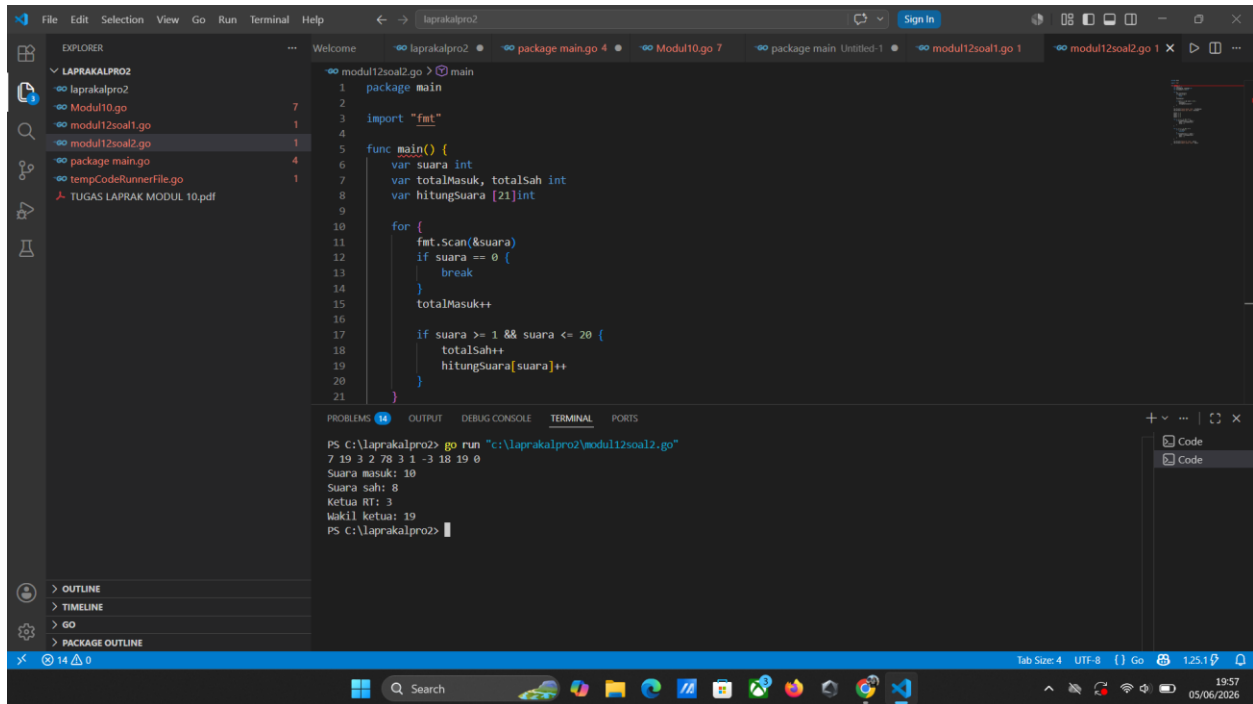
```
ketua := -1
wakil := -1
maks1 := -1
maks2 := -1

for i := 1; i <= 20; i++ {
    if hitungSuara[i] > maks1 {
        maks1 = hitungSuara[i]
        ketua = i
    }
}

for i := 1; i <= 20; i++ {
    if i == ketua {
        continue
    }
    if hitungSuara[i] > maks2 {
        maks2 = hitungSuara[i]
        wakil = i
    }
}

fmt.Printf("Ketua RT: %d\n", ketua)
fmt.Printf("Wakil ketua: %d\n", wakil)
}
```

screenshoot



Soal3

```
package main
```

```
import "fmt"
```

```
const NMAX = 1000000
```

```
var data [NMAX]int
```

```
func isiArray(n int) {
```

```
    for i := 0; i < n; i++ {
```

```

        fmt.Scan(&data[i])
    }
}

func posisi(n, k int) int {
    kiri := 0
    kanan := n - 1

    for kiri <= kanan {
        tengah := kiri + (kanan-kiri)/2
        if data[tengah] == k {
            return tengah
        } else if data[tengah] < k {
            kiri = tengah + 1
        } else {
            kanan = tengah - 1
        }
    }
    return -1
}

func main() {
    var n, k int
    fmt.Scan(&n, &k)

    isiArray(n)

```

```
hasil := posisi(n, k)
```

```
if hasil != -1 {
```

```
    fmt.Println(hasil)
```

```
} else {
```

```
    fmt.Println("TIDAK ADA")
```

```
}
```

```
}
```

screenshoot

